

Finite Element Analysis and Topology Optimization of an Automotive Engine Mount Bracket

Project Report

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Keywords: CAE, FEA, FEM, Automotive Engineering, Meshing, Static Structural Analysis, Von Mises Stress, Topology Optimization

Declaration

I hereby declare that this project report entitled “*Finite Element Analysis and Topology Optimization of an Automotive Engine Mount Bracket*” represents a technical study conducted for academic and engineering learning purposes.

The project methodology, calculations, software workflow, and simulation outputs are documented to the best of my knowledge. Solver-derived numerical results shall be reported only after verification using the corresponding CAE solver.

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Year: 2026

Acknowledgement

I would like to express my sincere gratitude to the faculty and mentors who supported the development of this CAE project.

I also acknowledge the importance of commercial CAE tools such as ANSYS and Altair HyperWorks in modern engineering simulation, as well as Python and OpenCascade for engineering workflow automation and geometry processing.

Abitha Chokka

Abstract

Automotive engine mounting systems are structural components responsible for transferring engine loads to the supporting vehicle structure while maintaining adequate stiffness and strength. Computer-Aided Engineering (CAE) provides a numerical approach for evaluating structural behavior before physical manufacturing.

This project presents a structured finite element analysis workflow for an automotive engine mount bracket. The methodology includes CAD geometry preparation, material definition, finite element mesh generation, boundary-condition definition, static structural analysis, topology optimization, and design validation.

The geometry workflow uses STEP-based CAD representation and OpenCascade-based geometry inspection. The mesh methodology considers tetrahedral solid elements, local refinement, mesh quality, and mesh convergence. Structural response is evaluated using quantities such as Von Mises stress, total deformation, and factor of safety.

A topology optimization stage is also incorporated to investigate potential material reduction while preserving structural requirements. The optimized geometry must subsequently be reconstructed, re-meshed, and re-analyzed under equivalent loading conditions.

The project provides a reproducible CAE workflow suitable for engineering analysis, student research, and portfolio demonstration.

Keywords: Finite Element Analysis, FEM, CAE, Engine Mount, Structural Analysis, Meshing, Von Mises Stress, Topology Optimization.

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Chapter 1

Introduction

1.1 Background

Automotive engine mounts are structural components used to connect the powertrain assembly to the supporting vehicle structure. They transfer mechanical loads and provide a controlled structural connection between the engine and vehicle body or frame.

The design of an engine mount must therefore consider strength, stiffness, load transfer, geometry, material properties, manufacturability, and durability.

Traditional design development can require repeated physical prototypes. CAE reduces the dependence on early-stage physical prototypes by allowing engineers to investigate structural response numerically.

1.2 Computer-Aided Engineering

Computer-Aided Engineering (CAE) uses computational methods to predict the behavior of engineering systems.

A typical structural CAE workflow is:

CAD → Geometry Cleanup → Material → Mesh → Boundary Conditions → Solver → Post-processing

1.3 Problem Statement

The objective of this project is to establish a systematic FEA workflow for studying an automotive engine mount bracket and to investigate whether topology optimization can reduce structural mass while satisfying defined structural requirements.

1.4 Project Objectives

1. Prepare and validate the CAD geometry.
2. Define appropriate material properties.
3. Develop an appropriate finite element mesh.
4. Perform mesh-quality and mesh-convergence studies.
5. Define realistic structural boundary conditions.
6. Evaluate static structural response.
7. Determine Von Mises stress and total deformation.
8. Calculate factor of safety from verified solver results.
9. Investigate topology optimization.
10. Compare original and optimized designs.

Chapter 2

Literature and Engineering Fundamentals

2.1 Finite Element Method

The Finite Element Method (FEM) divides a continuous physical domain into a finite number of smaller elements connected at nodes.

Each element approximates the behavior of the physical structure using interpolation or shape functions.

The general structural equilibrium equation is:

$$[K]\{u\} = \{F\}$$

where:

- $[K]$ = global stiffness matrix
- $\{u\}$ = displacement vector
- $\{F\}$ = applied force vector

2.2 Stress

Normal stress can be expressed as:

$$\sigma = \frac{F}{A}$$

where F is applied force and A is the resisting cross-sectional area.

2.3 Strain

Engineering strain is:

$$\epsilon = \frac{\Delta L}{L_0}$$

where ΔL is the change in length and L_0 is the original length.

2.4 Young's Modulus

For linear elastic behavior:

$$E = \frac{\sigma}{\epsilon}$$

where E is Young's modulus.

2.5 Poisson's Ratio

Poisson's ratio is defined as:

$$\nu = -\frac{\epsilon_{\text{lateral}}}{\epsilon_{\text{longitudinal}}}$$

2.6 Von Mises Stress

Von Mises stress is an equivalent stress measure commonly used in evaluating yielding of ductile materials.

For principal stresses σ_1 , σ_2 , and σ_3 :

$$\sigma_{vm} = \sqrt{\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}}$$

2.7 Factor of Safety

For a simple yield-based structural check:

$$FoS = \frac{\sigma_y}{\sigma_{vm,max}}$$

where:

- σ_y = material yield strength
- $\sigma_{vm,max}$ = maximum Von Mises stress

Chapter 3

Geometry Preparation

3.1 CAD Geometry

The automotive engine mount bracket is represented using the STEP neutral CAD format. The geometry contains a base mounting region and structural mounting ears with cylindrical bolt-hole features.

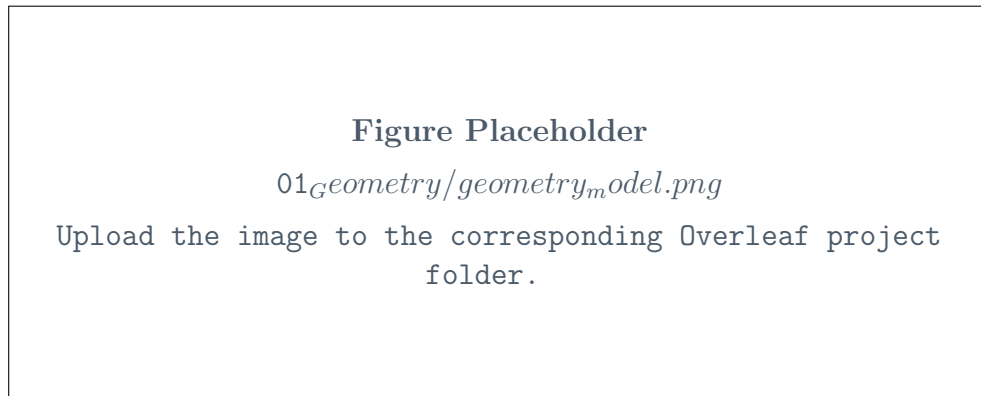


Figure 3.1: Automotive engine mount bracket CAD geometry.

3.2 Geometry Validation

The STEP model was inspected using OpenCascade-based geometry processing. The geometry-validation stage checks:

- Successful STEP import
- Solid topology
- Shell topology
- Surface faces
- Edges
- Vertices

The import process does not modify the original CAD model.

3.3 Geometry Cleanup Considerations

Before FEA, engineering geometry should be reviewed for:

- Small holes
- Sliver surfaces
- Very short edges
- Unnecessary cosmetic features
- Gaps or duplicate surfaces
- Sharp geometric transitions

Such features may unnecessarily increase mesh complexity without contributing significantly to the structural analysis.

Chapter 4

Material Definition

4.1 Material Properties

The structural material model requires the following properties:

Table 4.1: Material properties required for structural FEA

Property	Symbol	Unit
Young's Modulus	E	GPa
Poisson's Ratio	ν	–
Density	ρ	kg/m ³
Yield Strength	σ_y	MPa
Ultimate Strength	σ_u	MPa

Important: The final numerical material values should be populated from the verified material specification associated with the project.

4.2 Material Selection Considerations

Material selection influences:

- Structural stiffness
- Yield behavior
- Weight
- Stress distribution
- Safety factor
- Manufacturability

Chapter 5

Finite Element Meshing

5.1 Element Selection

The current project uses a three-dimensional solid-element strategy with tetrahedral elements.

A quadratic element formulation may be used where appropriate to improve geometric representation and displacement approximation.

5.2 Mesh Sizes

The mesh-convergence framework considers the following target element sizes:

Table 5.1: Planned mesh-convergence study

Mesh Level	Target Element Size
1	8 mm
2	6 mm
3	4 mm
4	3 mm
5	2 mm

5.3 Mesh Model

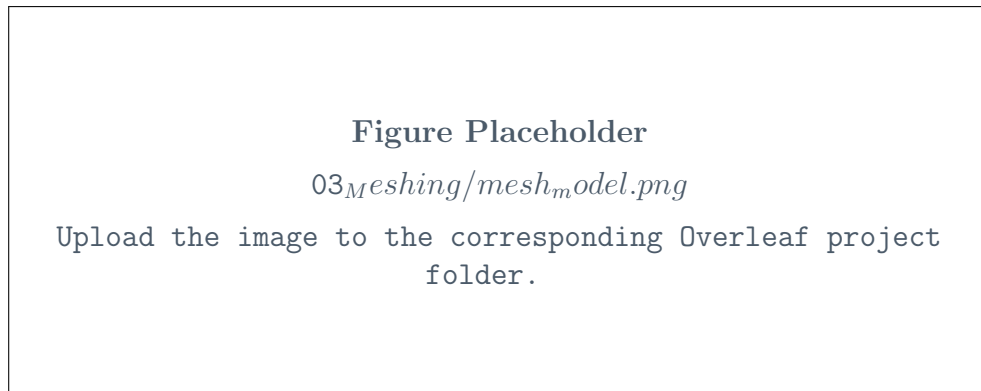


Figure 5.1: Finite element mesh of the engine mount bracket.

5.4 Mesh Quality

Important mesh-quality parameters include:

- Aspect ratio
- Skewness
- Jacobian
- Warpage
- Element angles
- Element quality

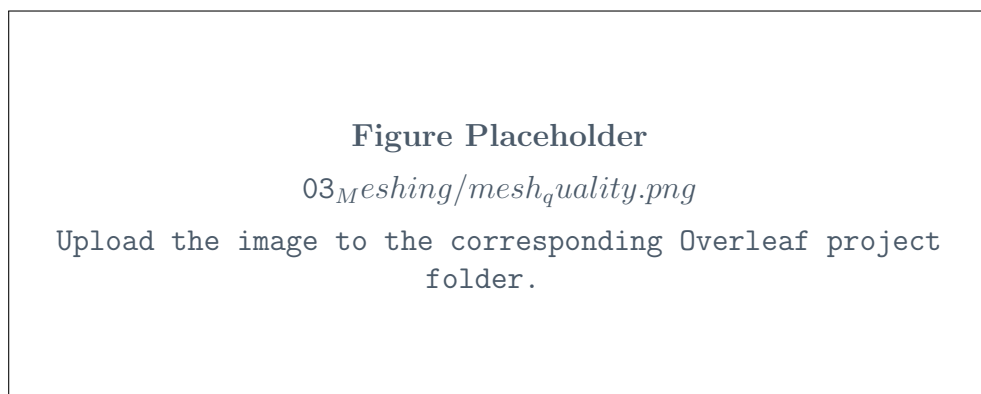


Figure 5.2: Mesh-quality inspection of the finite element model.

5.5 Local Refinement

Mesh refinement is particularly important in regions with high stress gradients, such as:

- Bolt holes
- Fillets
- Sharp transitions
- Load-transfer regions
- Fixed-support regions
- Contact regions

5.6 Mesh Convergence

Mesh convergence determines whether an important result has stabilized as the mesh is refined.

A representative convergence criterion is:

$$\% \Delta R = \left| \frac{R_i - R_{i-1}}{R_{i-1}} \right| \times 100$$

where R represents a selected response quantity such as maximum Von Mises stress or total deformation.

A predefined tolerance is then used to determine whether additional mesh refinement is necessary.

Chapter 6

Boundary Conditions and Loading

6.1 Fixed Support

The mounting regions connecting the bracket to the supporting structure are treated as constrained regions in the structural model.

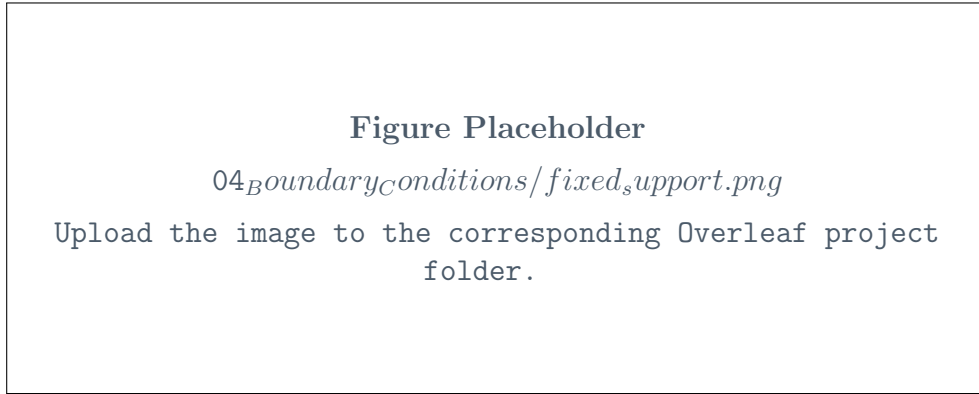


Figure 6.1: Fixed-support boundary condition applied to the mounting region.

6.2 Applied Load

A representative load is applied to the engine-mount load-transfer region.

The load direction is defined using the global coordinate system.

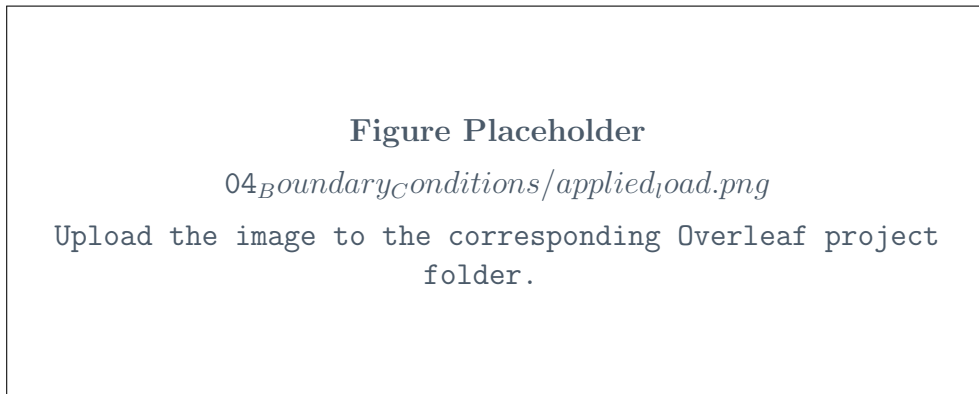


Figure 6.2: Applied structural load and global direction.

6.3 Boundary-Condition Considerations

Boundary conditions should represent the physical load-transfer mechanism as closely as practical.

An over-constrained model can artificially reduce deformation and alter stress distribution, while an under-constrained model may exhibit rigid-body motion.

Consequently, the support assumptions should be documented and reviewed before interpreting solver results.

Chapter 7

Static Structural Analysis

7.1 Analysis Type

A static structural analysis evaluates the structural response under loads that are sufficiently slow or steady that inertial effects can be neglected for the chosen model.

The principal outputs considered in this project are:

- Von Mises stress
- Total deformation
- Factor of safety

7.2 ANSYS / HyperWorks Workflow

The project workflow can be implemented using commercial CAE environments such as ANSYS Mechanical or Altair HyperWorks.

A typical sequence is:

1. Import geometry.
2. Clean and simplify geometry.
3. Define material.
4. Generate mesh.
5. Check mesh quality.
6. Apply boundary conditions.
7. Apply structural loads.
8. Solve.

9. Review stress and deformation.
10. Perform convergence and validation.

7.3 Von Mises Stress Result

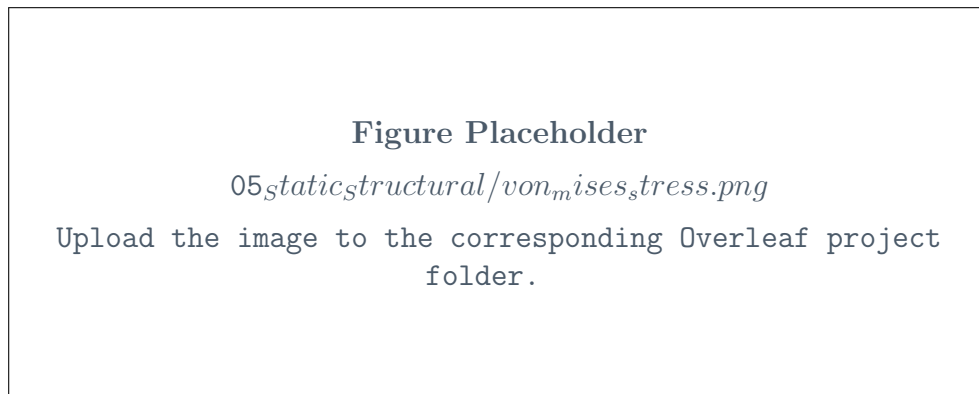


Figure 7.1: Von Mises stress contour obtained from the structural analysis.

7.4 Total Deformation Result

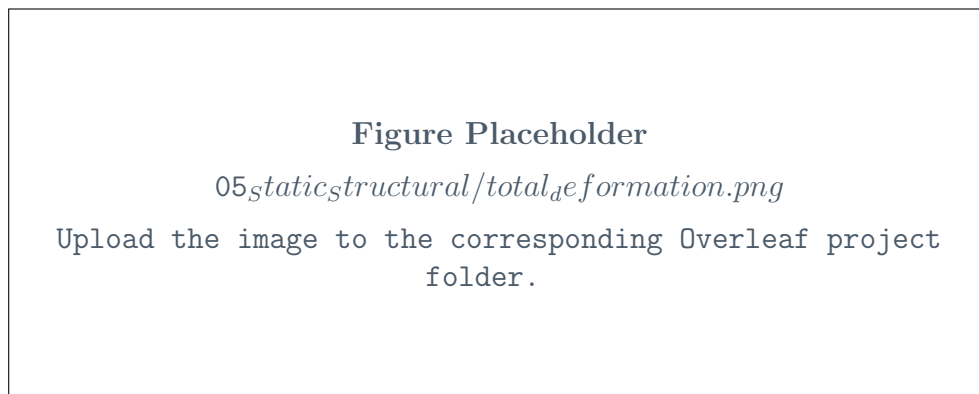


Figure 7.2: Total deformation contour obtained from the structural analysis.

7.5 Factor of Safety

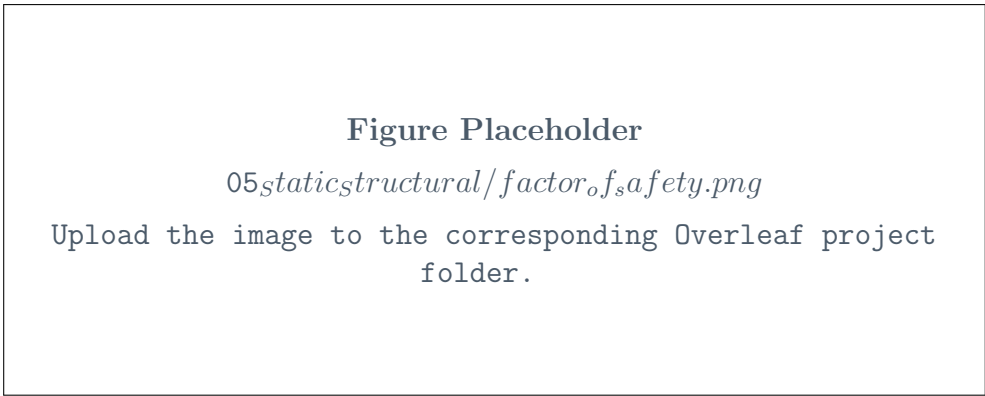


Figure 7.3: Factor-of-safety contour or post-processed structural result.

7.6 Result Table

Table 7.1: Static structural analysis results

Parameter	Value	Unit
Maximum Von Mises Stress	To be updated	MPa
Maximum Total Deformation	To be updated	mm
Yield Strength	To be updated	MPa
Factor of Safety	To be calculated	–

Note: Numerical values in this table should be populated only from verified solver output.

Chapter 8

Topology Optimization

8.1 Purpose

Topology optimization investigates the distribution of material within a specified design domain while satisfying defined structural requirements.

The objective in this project is to investigate potential structural mass reduction.

8.2 Optimization Definition

Table 8.1: Topology optimization formulation

Parameter	Definition
Design variable	Material distribution within the design domain
Objective	Potential reduction in structural mass
Constraints	Structural performance requirements
Non-design regions	Mounting and critical load-transfer regions
Solver	ANSYS or Altair OptiStruct

8.3 Optimization Setup

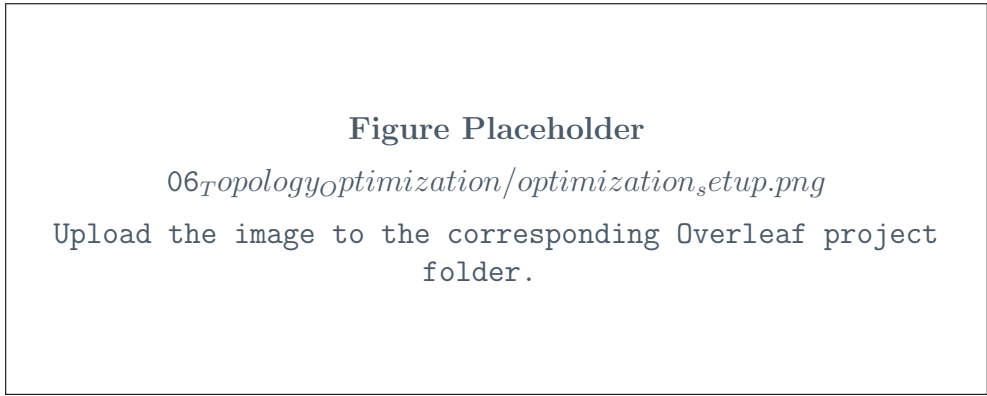


Figure 8.1: Topology-optimization design-space and constraint definition.

8.4 Optimized Design

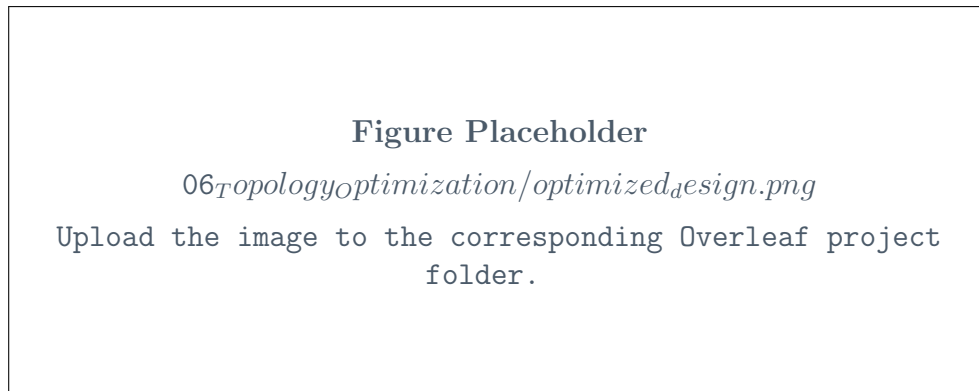


Figure 8.2: Optimized engine-mount bracket geometry.

8.5 Post-Optimization Re-analysis

The optimized topology should not be considered the final engineering design by optimization alone.

The recommended workflow is:

Optimized Topology → Reconstructed CAD → Re-meshing → Structural Re-analysis → Validation

Chapter 9

Validation and Comparison

9.1 Validation Strategy

Validation compares the original and optimized configurations under equivalent material and structural loading conditions.

Important comparison parameters are:

- Mass
- Von Mises stress
- Total deformation
- Factor of safety
- Mesh quality
- Relevant loading cases

9.2 Original vs Optimized Comparison

Table 9.1: Comparison of original and optimized designs

Metric	Original	Optimized	Change
Mass	To update	To update	To update
Von Mises Stress	To update	To update	To update
Total Deformation	To update	To update	To update
Factor of Safety	To update	To update	To update

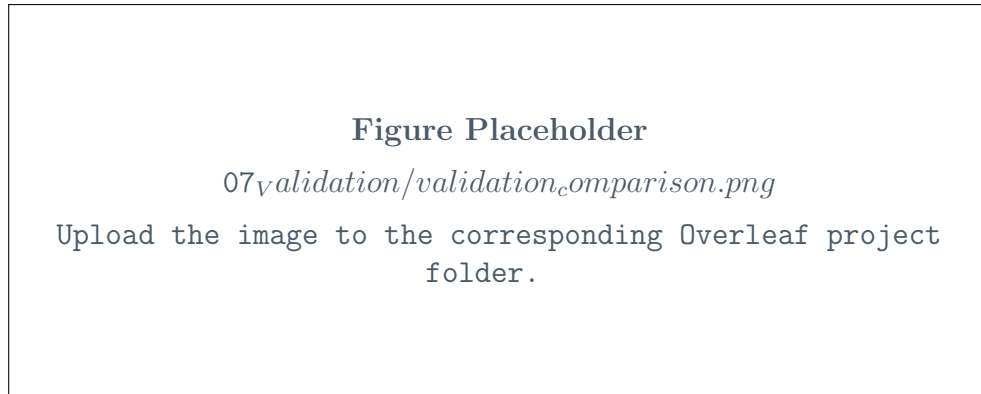


Figure 9.1: Original versus optimized design comparison.

9.3 Engineering Interpretation

The optimized design should be evaluated not only for mass reduction but also for the resulting stress and deformation response.

A useful optimization result is therefore a design that satisfies its defined structural requirements while achieving the intended reduction in material usage.

Manufacturing constraints and geometric reconstruction should also be considered before treating an optimized topology as a manufacturable component.

Chapter 10

Discussion

10.1 CAE Methodology

The project demonstrates a complete structural CAE workflow beginning from CAD geometry and progressing through meshing, boundary-condition definition, structural analysis, optimization, and validation.

The separation between model preparation and solver post-processing allows the workflow to be documented and reproduced systematically.

10.2 Mesh Considerations

Mesh density strongly influences computational cost and the accuracy of local stress predictions.

A globally fine mesh is not always efficient. Local refinement around high-gradient regions can provide improved resolution while controlling computational cost.

Mesh convergence should therefore be demonstrated using actual solver results rather than assuming that a particular element size is automatically sufficient.

10.3 Stress Concentrations

Engine mounting brackets can contain geometric stress concentrations near holes, fillets, sharp transitions, and load-transfer regions.

Very localized peak values should therefore be interpreted carefully, especially where stress singularities or idealized boundary conditions may exist.

10.4 Topology Optimization Considerations

Topology optimization can generate geometrically complex material distributions.

The optimized result should subsequently be transformed into a manufacturable CAD design and re-evaluated using a new mesh.

Chapter 11

Conclusion

This project establishes a structured CAE methodology for the analysis and optimization of an automotive engine mount bracket.

The workflow incorporates CAD geometry preparation, material definition, finite element meshing, boundary conditions, static structural analysis, topology optimization, and validation.

The methodology emphasizes mesh quality and convergence, appropriate boundary-condition definition, interpretation of Von Mises stress and deformation, and verification of the optimized design through re-analysis.

Future work includes completing the model using verified material data, obtaining solver-generated structural results, performing a quantitative mesh-convergence study, executing topology optimization in ANSYS or Altair OptiStruct, and validating the reconstructed optimized geometry.

Chapter 12

Future Scope

Future development of this project may include:

1. Multiple engine load cases.
2. Modal analysis.
3. Harmonic response analysis.
4. Fatigue analysis.
5. Buckling analysis.
6. Temperature-dependent structural analysis.
7. Nonlinear contact analysis.
8. Manufacturing-constrained topology optimization.
9. Experimental validation using physical prototypes.

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6. OpenCascade Technology Documentation, <https://dev.opencascade.org/>

Chapter A

Project File Structure

Automotive_Engine_Mount_FEA/

01_Geometry/

engine_mount_bracket.step

geometry_import.py

02_Material/

material_properties.pdf

material_properties.py

03_Meshing/

mesh_setup.py

mesh_model.png

mesh_quality.png

04_Boundary_Conditions/

fixed_support.png

applied_load.png

load_setup.py

05_Static_Structural/

von_mises_stress.png

total_deformation.png

factor_of_safety.png

static_structural_analysis.py

06_Topology_Optimization/

optimization_setup.png

optimized_design.png

topology_optimization.py

07_Validation/

validation.py

original_vs_optimized.csv

Chapter B

Important Engineering Data

Table B.1: Project data status

Item	Status	Source
CAD geometry	Available	STEP model
Material properties	To verify	Material specification
Mesh	Framework prepared	CAE solver
Boundary conditions	Defined	Engineering model
Static results	Pending	ANSYS / HyperWorks
Topology results	Pending	ANSYS / OptiStruct
Validation	Pending	Solver comparison

Chapter C

Key Equations

$$\sigma = \frac{F}{A}$$

$$\epsilon = \frac{\Delta L}{L_0}$$

$$E = \frac{\sigma}{\epsilon}$$

$$FoS = \frac{\sigma_y}{\sigma_{vm,\max}}$$

$$\% \Delta R = \left| \frac{R_i - R_{i-1}}{R_{i-1}} \right| \times 100$$